Test Case / Original Request

Run PTC-005 using K6. Execute the real performance test and collect monitoring results. Assess the security surface for the same job and exact configured target.Analyze the current job’s results and generate the final PDF with security coverage and evidence, plus links to the native JMeter report, raw results, and final PDF.

REAL Execution Outcome

Execution status: COMPLETED
Runner status: completed
Outcome details: REAL k6 execution completed. Threshold breaches, if any, are retained as performance findings.
Thresholds passed: True

Executive Summary

Application: Unknown
Scenario: Task Status API — Load Check
Job ID: JOB-9907ffe5f5df406586fb1ad68aa299ab
Overall Result: WARNING

Performance testing revealed a significant latency outlier for /todos/1, with the max response time exceeding the p95 by more than 11x, and warning-level post-ramp latency degradation. No errors were observed but the load-generator host memory was critically high. Security assessment returned a WARNING: Strict-Transport-Security was missing in the observed API response (single HEAD request), so HSTS control effectiveness is not established. No historical comparison is available for anomaly context.

Execution Profile and Artifacts

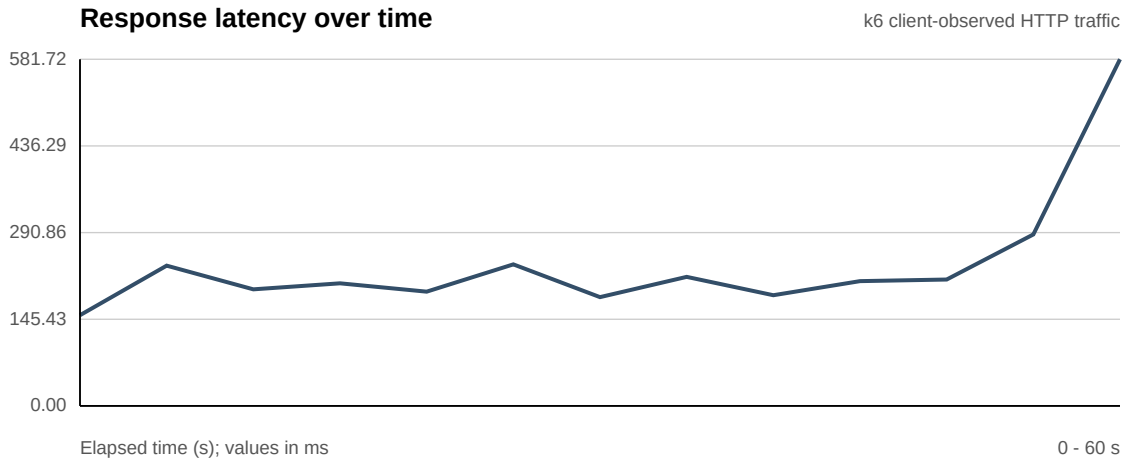
Execution mode: REAL
Workload profile:
- tool: k6
- target_url: https://jsonplaceholder.typicode.com
- endpoint: /todos/1
- method: GET
- users: 25
- max_rps: 15
- duration_seconds: 60
- ramp_up_seconds: 5
- expected_status_code: 200
- p95_threshold_ms: 5000
Artifact directory: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab
Artifact references:
- aligned_time_series: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check_aligned_time_series.csv
- generated_script: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check.js
- k6_summary: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check_summary.json

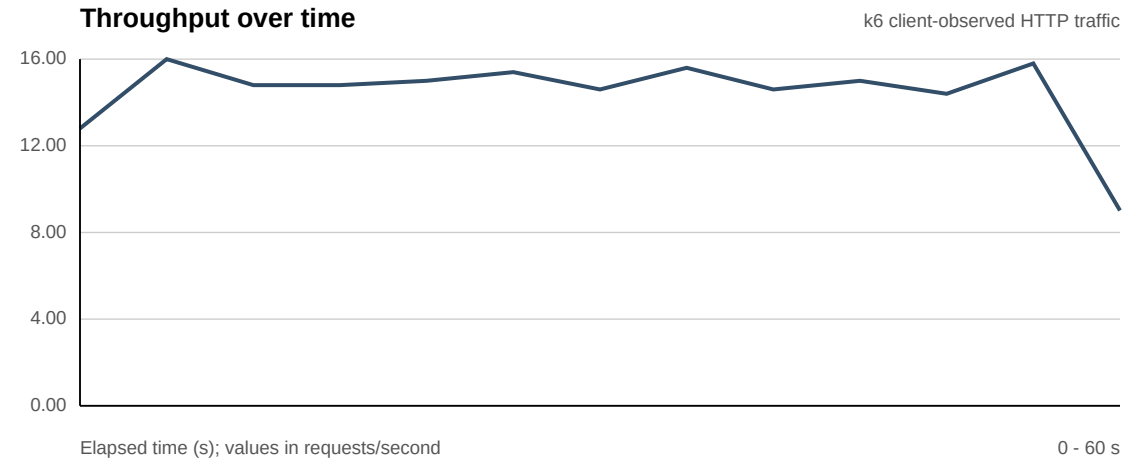
- measurement_metadata: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check_measurement_metadata.json
- monitoring_samples: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check_runtime_samples.csv
- Native k6 HTML report
- Native report + raw results ZIP
- normalized_metrics: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check_normalized_metrics.json
- Raw k6 results (JSON)
- runner_log: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check_runner.log
- time_series: src/generated_scripts/k6 native reports/JOB-9907ffe5f5df406586fb1ad68aa299ab/k6_PTC_005_Task_Status_API_Load_Check_time_series.csv

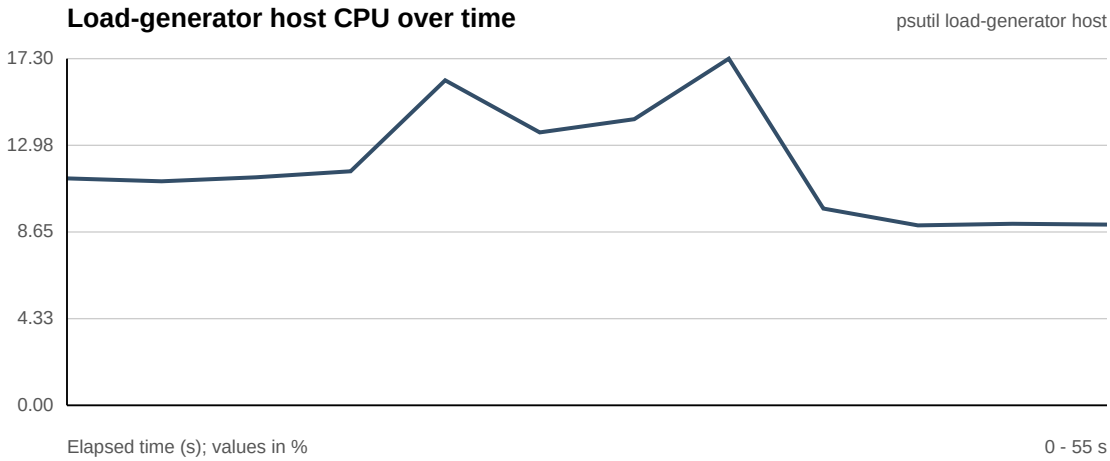
The interactive native HTML dashboard is linked as a separate artifact and is not embedded in this PDF.

Time-Series Charts (REAL Measurements)

Charts show measured values in five-second aligned buckets. Host CPU is from the load-generator machine, not the target.







Agent Activity Summary

Agent	*Summary*
Test Execution	Load test executed on /todos/1 (Unknown - Task Status API — Load Check). Results: 896 total requests, 896 successful, 0 failed. Avg response time: 211.61572767857146 ms; 95th percentile: 339.92374999999987 ms; Max: 3857.206 ms. Throughput: 14.891896135938547 TPS; Error rate: 0.0%.
Monitoring	CPU usage: 12.0% average, 27.3% peak; memory: 79.87% average, 81.7% peak; DB connections: not collected, TPS: 14.891896135938547, Response time: 211.61572767857146 ms. Data source and scope: REAL: psutil load-generator host samples and k6 client-observed metrics. Measurement provenance: cpu_usage=psutil (load_generator_host); cpu_peak_usage=psutil (load_generator_host); memory_usage=psutil (load_generator_host); memory_peak_usage=psutil (load_generator_host); response_time=k6 (load_generator_observed_http_traffic); tps=k6 (load_generator_observed_http_traffic); db_connections=not_collected (No target-specific database telemetry provider is configured.). Alerts triggered: ['High load-generator host memory usage']

Agent	*Summary*
Security Validation	Status: WARNING. Security score: 0%. Live security check: completed. Issue: Strict-Transport-Security was not present in the observed response; this single response does not establish the full origin policy. Issue: Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness. Issue: Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness. Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0} Control assessment evidence: - external_surface: assessed; basis=live_observation; evidence=['live-observation:JOB-9907ffe5f5df406586fb1ad68aa299ab'] Evidence references: live-observation:JOB-9907ffe5f5df406586fb1ad68aa299ab Recommendation: Review whether an HSTS policy is appropriate for this HTTPS API origin.
Analysis	Performance testing revealed a significant latency outlier for /todos/1, with the max response time exceeding the p95 by more than 11x, and warning-level post-ramp latency degradation. No errors were observed but the load-generator host memory was critically high. Security assessment returned a WARNING: Strict-Transport-Security was missing in the observed API response (single HEAD request), so HSTS control effectiveness is not established. No historical comparison is available for anomaly context.
Reporting	A PDF report was generated, including all findings, recommendations, and an executive summary.

Per-API Findings (Test Execution)

- /todos/1 [WARNING / medium]
avg=211.61572767857146ms p95=339.92374999999987ms max=3857.206ms throughput=14.891896135938547 req/s error_rate=0.0% http_status=200
* Max response time (3857.206ms) is 11.3x p95 (339.92374999999987ms) -- outlier spikes present.
history: Historical context is unavailable (RAG not accessible); analysis is based on current run threshold breaches only.
recommendation: This API exhibited significant response time spikes, per breach of the max-to-p95 ratio threshold. No additional historical insight or runbook references are available due to missing context; focus on investigating possible causes for transient spikes and outliers.

Infrastructure Health (Monitoring)

Status: WARNING (monitoring_agent reported: WARNING)
CPU usage: 12.0% Memory usage: 79.87% DB connections: None TPS: 14.891896135938547
Response time (monitoring): 211.61572767857146ms
- 1 monitoring alert(s) reported: ['High load-generator host memory usage']
Monitoring alerts: ['High load-generator host memory usage']

Security Findings (Security Agent)

Status: WARNING Security score: 0%
Security result job ID: JOB-9907ffe5f5df406586fb1ad68aa299ab.
Live security check: completed.
Issues:
- Strict-Transport-Security was not present in the observed response; this single response does not establish the full origin policy.
- Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness.
- Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness.
Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0}
Requested controls: external_surface
Assessed controls: external_surface
Control evidence and provenance:
- external_surface: assessed; basis=live_observation; evidence fields=['target_origin', 'endpoint', 'method', 'response_status_code', 'observed_security_headers']; references=['live-observation:JOB-9907ffe5f5df406586fb1ad68aa299ab']
Evidence references: live-observation:JOB-9907ffe5f5df406586fb1ad68aa299ab
Live observation: HEAD /todos/1; HTTP 200; elapsed 909.1651249909773 ms; requests sent 1.
Observed response headers (observations only):
- cache-control: max-age=43200
- x-content-type-options: nosniff
Recommendations:
- Review whether an HSTS policy is appropriate for this HTTPS API origin.

Recommended Next Steps

- /todos/1: This API exhibited significant response time spikes, per breach of the max-to-p95 ratio threshold. No additional historical insight or runbook references are available due to missing context; focus on investigating possible causes for transient spikes and outliers.
- Infrastructure: review CPU/memory/DB connection anomalies above.
- Security: address the issues/recommendations listed above before next deployment.